

Notes: Langlois (JFE, 2023)

What matters in a characteristic?

Contents

1	Big picture	2
2	Data and measurement	2
3	Models and empirical specifications	2
3.1	Characteristic decomposition	2
3.2	Latent factor return model	2
3.3	Total and Predictive R^2	2
3.4	Alpha and component-group tests	3
4	Identification and empirical design	3
5	Tables and figures	3
5.1	Table 1: Cross-sectional regression for characteristics – Average R^2	3
5.2	Table 2: Model performance – Total R^2	4
5.3	Table 3: Model performance – Predictive R^2	4
5.4	Table 4: Model performance – Asset pricing tests	4
5.5	Table 5 and Table 6: Comparison to other factor models; Significance tests for parameter groups	4
5.6	Figure 1: What is the relative importance of each characteristic component for alphas and betas?	5
5.7	Figure 2: Are alphas significant over time?	5
5.8	Figure 3: Component contributions to alphas and beta over time	6
5.9	Figure 4: Risk-adjusted returns of high-minus-low predicted alpha portfolios	6
5.10	Figure 5: Risk-adjusted returns of high predicted alpha portfolios	6
6	Short summary	7
7	Conclusion	7
7.1	Core conclusion	7
7.2	Economic intuition	7
7.3	Implications for diversification	7
7.4	Implications for anomaly investing	8
7.5	Limitations and future research	8
7.6	Takeaway	8

1 Big picture

- **Question.** Which part of a stock characteristic matters for expected returns and which part matters for return comovement: country, industry, or country-and-industry-adjusted variation?
- **Core idea.** Firm characteristics are decomposed into country components (CtyX), industry components (IndX), and adjusted residual components (AdjX).
- **Methodological contribution.** The paper combines characteristic decomposition with IPCA, so characteristic components instrument both alphas and factor loadings.
- **Main alpha result.** Adjusted components are the dominant source of alphas in international markets.
- **Main beta result.** Country components dominate factor exposures and comovements.
- **Trading result.** Statistically significant alphas are not robustly tradable after estimation risk and transaction costs.

2 Data and measurement

The sample covers 55,541 stocks across 35 countries and 17 industries. The paper estimates models for U.S., developed markets excluding the U.S., emerging markets, and the World sample. The eight characteristics are size, value, profitability, investment, momentum, lagged return, market beta, and idiosyncratic volatility. All characteristics are transformed into normalized cross-sectional ranks.

Interpretation. The regional split is essential because U.S. results differ from international results.

3 Models and empirical specifications

3.1 Characteristic decomposition

$$x_{n,t}^{(j)} = \kappa_t^{(j)} + \sum_c C_{c,t}^{(j)} \mathbf{1}\{n \in c\} + \sum_i I_{i,t}^{(j)} \mathbf{1}\{n \in i\} + v_{n,t}^{(j)}.$$

Interpretation. The residual v is the adjusted characteristic component, i.e., the within-country and within-industry part of the characteristic.

3.2 Latent factor return model

$$r_{n,t+1} = \alpha_{n,t} + \beta'_{n,t} f_{t+1} + \epsilon_{n,t+1}, \quad \alpha_{n,t} = A' z_{n,t} + \nu_{\alpha,n,t}, \quad \beta_{n,t} = B' z_{n,t} + \nu_{\beta,n,t}.$$

Interpretation. The same instruments can explain alphas and betas. This allows the model to distinguish expected-return effects from risk-exposure effects.

3.3 Total and Predictive R^2

$$\text{Total } R^2 = 1 - \frac{\sum_{n,t} (r_{n,t+1} - \alpha_{n,t} - \beta'_{n,t} f_{t+1})^2}{\sum_{n,t} r_{n,t+1}^2}.$$

$$\text{Predictive } R^2 = 1 - \frac{\sum_{n,t} (r_{n,t+1} - \alpha_{n,t} - \beta'_{n,t} \lambda)^2}{\sum_{n,t} r_{n,t+1}^2}.$$

Interpretation. Total R^2 captures realized return comovement. Predictive R^2 captures conditional expected-return variation.

3.4 Alpha and component-group tests

$$H_0 : A' A = 0, \quad H_0 : A'_c A_c = 0, \quad H_0 : A'_i A_i = 0, \quad H_0 : A'_a A_a = 0.$$

For beta contributions, the paper uses $\text{tr}(B_j S_f B'_j)$ for $j \in \{c, i, a\}$.

Interpretation. These statistics show which component families matter for alpha and beta parameters.

4 Identification and empirical design

The paper is a measurement and asset-pricing decomposition exercise rather than a causal identification design. Its credibility comes from decomposing characteristics every month, using IPCA to let factors be latent, imposing the $A'B = 0$ restriction so risk receives explanatory priority, comparing many model specifications, and validating the results through rolling-window and trading tests.

5 Tables and figures

5.1 Table 1: Cross-sectional regression for characteristics – Average R^2

Read: Country and industry affiliation explain substantial characteristic variation, especially in EM. In the World sample, the combined average R^2 is 12.63%; in EM it is 19.40%.

Economic message. This motivates the decomposition because unadjusted characteristics contain meaningful country and industry structure.

Table 1
Cross-sectional regression for characteristics – Average R^2 .

Characteristic	Industry dummies	Country dummies	Industry and country dummies	Industry dummies	Country dummies	Industry and country dummies
<i>Panel A: U.S.</i>						
Size	9.08	-	-	6.80	7.17	12.72
Value	4.13	-	-	7.44	10.38	16.14
Profitability	12.48	-	-	10.96	21.75	29.29
Investment	2.38	-	-	4.41	5.65	7.94
Momentum	3.45	-	-	3.90	6.62	9.59
Lagged return	2.70	-	-	2.92	7.25	9.19
Beta	8.31	-	-	5.95	4.61	9.98
Idio-vol	7.31	-	-	7.06	8.38	12.51
Average	6.23	-	-	6.18	8.98	13.42
<i>Panel B: DM ex U.S.</i>						
Size	6.51	29.83	33.76	5.49	17.95	21.36
Value	3.79	27.40	30.18	5.18	16.13	20.55
Profitability	6.87	12.71	18.91	4.41	9.27	12.54
Investment	0.79	8.03	8.83	2.25	4.81	6.34
Momentum	1.68	15.23	16.52	2.29	8.55	10.28
Lagged return	1.50	17.64	18.41	1.84	9.45	10.71
Beta	3.08	10.65	13.28	4.83	5.40	10.57
Idio-vol	1.85	13.86	15.48	3.57	5.57	8.72
Average	3.26	16.92	19.40	3.73	9.64	12.63
<i>Panel C: EM</i>						
Size	6.51	29.83	33.76	5.49	17.95	21.36
Value	3.79	27.40	30.18	5.18	16.13	20.55
Profitability	6.87	12.71	18.91	4.41	9.27	12.54
Investment	0.79	8.03	8.83	2.25	4.81	6.34
Momentum	1.68	15.23	16.52	2.29	8.55	10.28
Lagged return	1.50	17.64	18.41	1.84	9.45	10.71
Beta	3.08	10.65	13.28	4.83	5.40	10.57
Idio-vol	1.85	13.86	15.48	3.57	5.57	8.72
Average	3.26	16.92	19.40	3.73	9.64	12.63
<i>Panel D: World</i>						
Size	6.51	29.83	33.76	5.49	17.95	21.36
Value	3.79	27.40	30.18	5.18	16.13	20.55
Profitability	6.87	12.71	18.91	4.41	9.27	12.54
Investment	0.79	8.03	8.83	2.25	4.81	6.34
Momentum	1.68	15.23	16.52	2.29	8.55	10.28
Lagged return	1.50	17.64	18.41	1.84	9.45	10.71
Beta	3.08	10.65	13.28	4.83	5.40	10.57
Idio-vol	1.85	13.86	15.48	3.57	5.57	8.72
Average	3.26	16.92	19.40	3.73	9.64	12.63

We report the time-series average adjusted R^2 (in %) for the cross-sectional regressions in Eq. (1). Each month, we run a cross-sectional regression for each characteristic of their values on either industry dummy variables, country variables, or both industry and country dummy variables. We run regressions using all stocks in the U.S. in Panel A, for which specifications with country dummy variables are not estimated, in Developed Markets excluding the U.S. in Panel B, in Emerging Markets in Panel C, and in all countries in Panel D. We use a 17-industry classification. All data are monthly, in USD, and end in

Table 2
Model performance – Total R^2 .

Model	L	Number K of factors							
		1	2	3	4	5	6	7	8
<i>Panel A: United States of America, 1988–2022</i>									
Unadj X	9	11.99							
Ind and adj X	17	11.99	13.14	13.99	14.41	14.64	14.84	15.01	15.11
Ind X	9	10.27	10.88	11.18	11.35	11.54	11.65	11.74	11.82
Ind D	17	10.10	10.95	11.46	11.67	11.85	11.96	12.06	12.13
<i>Panel B: Developed Markets ex U.S., 1991–2022</i>									
Unadj X	9	11.57	12.60	13.13	13.59	13.95	14.25	14.47	14.57
Ind and adj X	17	11.64	12.73	13.37	13.85	14.24	14.56	14.84	15.03
Ind X	9	10.62	11.10	11.51	11.75	11.90	12.02	12.10	12.17
Ind D	17	10.53	11.50	11.96	12.27	12.41	12.48	12.55	12.60
Cty and adj X	17	11.74	13.81	14.81	15.41	15.89	16.22	16.55	16.82
Cty X	9	10.72	12.76	13.51	14.01	14.29	14.52	14.68	14.82
Cty D	15	10.39	13.53	14.26	14.70	14.90	15.05	15.15	15.23
Cty, ind, and adj X	25	11.82	13.92	14.89	15.52	16.07	16.44	16.76	17.04
Cty and ind X	17	11.02	13.11	13.95	14.42	14.76	15.03	15.25	15.42
Cty and ind D	31	10.66	13.78	14.77	15.25	15.67	16.00	16.17	16.32
<i>Panel C: Emerging Markets, 1999–2022</i>									
Unadj X	9	13.25	15.54	16.43	17.02	17.37	17.67	17.85	18.00
Ind and adj X	17	13.28	15.60	16.50	17.09	17.43	17.78	17.99	18.15
Ind X	9	12.63	12.85	13.10	13.22	13.31	13.38	13.44	13.49
Ind D	17	12.62	13.09	13.26	13.39	13.50	13.56	13.61	13.65
Cty and adj X	17	13.47	18.24	19.91	21.11	21.95	22.61	23.13	23.58
Cty X	9	12.91	17.67	19.31	20.51	21.34	21.98	22.49	22.94
Cty D	19	13.16	18.89	21.54	22.62	23.45	24.04	24.53	24.96
Cty, ind, and adj X	25	13.51	18.22	19.98	21.23	22.05	22.74	23.27	23.75
Cty and ind X	17	12.99	17.70	19.42	20.67	21.48	22.15	22.68	23.15
Cty and ind D	35	13.22	18.96	21.62	22.71	23.55	24.15	24.66	25.08
<i>Panel D: World, 1999–2022</i>									
Unadj X	9	10.42	11.30	11.89	12.34	12.68	12.86	13.01	13.14
Ind and adj X	17	10.51	11.44	12.06	12.53	12.90	13.11	13.27	13.43
Ind X	9	9.58	9.89	10.09	10.24	10.34	10.42	10.49	10.53
Ind D	17	9.49	9.97	10.23	10.48	10.61	10.72	10.77	10.81
Cty and adj X	17	10.63	12.15	13.27	14.02	14.61	15.07	15.49	15.85
Cty X	9	9.67	11.15	12.18	12.88	13.39	13.79	14.11	14.37
Cty D	35	9.84	12.14	13.86	14.62	15.22	15.66	16.00	16.29
Cty, ind, and adj X	25	10.74	12.27	13.39	14.19	14.84	15.31	15.74	16.10
Cty and ind X	17	9.96	11.45	12.54	13.23	13.79	14.24	14.57	14.85
Cty and ind D	51	10.04	12.37	14.13	14.94	15.57	16.08	16.45	16.79

We report the Total R^2 's from Eq. (5) in percent for different models and numbers of factors. Models are estimated on all stocks in the U.S. in Panel A, for which models with country effects are not estimated, in Developed Markets excluding the U.S. in Panel B, in Emerging Markets in Panel C, and in all countries in Panel D. Models differ by the instrument used in the specification of the u 's (see Eq. (3)) and β 's (see Eq. (4)). Model "rhadj X" uses unadjusted characteristics. Model "ind and adj X" uses industry and industry-adjusted characteristic components, while Model "ind X" uses only industry components. Model "ind D" uses industry dummies. Model "Cty and adj X" uses country and country-adjusted characteristic components, while Model "Cty X" uses only country components. Model "Cty D" uses country dummies. Model "Cty, ind, and adj X" uses country, industry, and country- and industry-adjusted characteristic components, while Model "Cty and ind X" uses country and industry components. Finally, Model "Cty and ind D" uses country and industry

5.2 Table 2: Model performance – Total R^2

Read: Component-based models generally have the highest Total R^2 . In the U.S., “Ind and adj X” reaches 15.57% with $K = 8$. In DM ex U.S., “Cty, ind, and adj X” reaches 17.04%.

Economic message. Country and adjusted components improve the model’s ability to capture comovement.

5.3 Table 3: Model performance – Predictive R^2

Read: Predictive R^2 values are small, but models containing adjusted components perform best or near-best across regions.

Economic message. The adjusted component provides expected-return content that dummy models and country-only models miss.

Table 3
Model performance – Predictive R^2 .

Model	L	Number K of factors							
		1	2	3	4	5	6	7	8
Panel A: United States of America									
Unadj X	9	0.77	0.76	0.76	0.76	0.76	0.76	0.76	0.76
Ind and adj X	17	0.88	0.85	0.85	0.86	0.86	0.86	0.85	0.84
Ind X	9	0.66	0.64	0.65	0.64	0.65	0.63	0.63	0.62
Ind D	17	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55
Panel B: Developed Markets ex U.S.									
Unadj X	9	0.30	0.29	0.29	0.29	0.29	0.29	0.29	0.29
Ind and adj X	17	0.47	0.45	0.44	0.44	0.44	0.43	0.44	0.44
Ind X	9	0.34	0.34	0.33	0.33	0.32	0.32	0.32	0.32
Ind D	17	0.18	0.18	0.18	0.18	0.18	0.18	0.18	0.18
Cty and adj X	17	0.31	0.30	0.30	0.30	0.30	0.30	0.29	0.29
Cty X	9	0.20	0.20	0.19	0.19	0.19	0.19	0.18	0.19
Cty D	15	0.19	0.18	0.18	0.18	0.18	0.18	0.18	0.18
Cty, ind, and adj X	25	0.45	0.44	0.44	0.42	0.41	0.42	0.42	0.42
Cty and ind X	17	0.33	0.32	0.31	0.31	0.31	0.30	0.31	0.31
Cty and ind D	31	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.19
Panel C: Emerging Markets									
Unadj X	9	0.65	0.65	0.64	0.64	0.64	0.64	0.64	0.65
Ind and adj X	17	0.73	0.72	0.72	0.72	0.72	0.72	0.72	0.72
Ind X	9	0.58	0.58	0.58	0.57	0.57	0.57	0.57	0.57
Ind D	17	0.51	0.51	0.51	0.51	0.51	0.51	0.51	0.51
Cty and adj X	17	0.76	0.77	0.77	0.79	0.75	0.74	0.70	0.71
Cty X	9	0.59	0.60	0.59	0.61	0.57	0.56	0.58	0.52
Cty D	19	0.53	0.52	0.51	0.52	0.52	0.52	0.52	0.52
Cty, ind, and adj X	25	0.82	0.80	0.83	0.84	0.81	0.81	0.78	0.80
Cty and ind X	17	0.64	0.65	0.62	0.67	0.64	0.63	0.61	0.63
Cty and ind D	35	0.54	0.53	0.51	0.52	0.52	0.52	0.52	0.52
Panel D: World									
Unadj X	9	0.50	0.50	0.50	0.49	0.49	0.49	0.49	0.49
Ind and adj X	17	0.60	0.59	0.59	0.58	0.57	0.57	0.57	0.57
Ind X	9	0.47	0.47	0.46	0.45	0.45	0.44	0.44	0.44
Ind D	17	0.38	0.38	0.38	0.38	0.38	0.38	0.38	0.38
Cty and adj X	17	0.52	0.51	0.50	0.53	0.52	0.53	0.53	0.53
Cty X	9	0.40	0.38	0.38	0.39	0.39	0.40	0.39	0.40
Cty D	35	0.41	0.40	0.39	0.40	0.41	0.39	0.39	0.41
Cty, ind, and adj X	25	0.63	0.61	0.62	0.62	0.62	0.60	0.60	0.60
Cty and ind X	17	0.50	0.48	0.48	0.49	0.48	0.49	0.49	0.48
Cty and ind D	51	0.42	0.40	0.40	0.42	0.41	0.41	0.41	0.41

We report the Predictive R^2 from Eq. (5) in percent for different models and numbers of factors. Models are estimated on all stocks in the U.S. in Panel A, for which models with country effects are not estimated, in Developed Markets excluding the U.S. in Panel B, in Emerging Markets in Panel C, and in all countries in Panel D. Models differ by the instrument used in the specification of the α s (see Eq. (3)) and β s (see Eq. (4)). Model “Unadj X” uses unadjusted characteristics. Model “Ind and adj X” uses industry and industry-adjusted characteristic components, while Model “Ind X” uses only industry components. Model “Ind D” uses industry dummies. Model “Cty and adj X” uses country and country-adjusted characteristic components, while Model “Cty X” uses only country components. Model “Cty D” uses country dummies. Model “Cty, ind, and adj X” uses country, industry, and country- and industry-adjusted characteristic components, while Model “Cty and ind X” uses country and industry components. Finally Model “Cty and ind D” uses country and industry dummies.

Table 4
Model performance – Asset pricing tests.

Model	L	Number K of factors							
		1	2	3	4	5	6	7	8
Panel A: United States of America									
Unadj X	9	0.00	0.00	0.00	0.16	0.00	54.74	25.14	1.64
Ind and adj X	17	0.06	0.00	0.00	0.00	0.00	0.00	0.00	3.30
Ind X	9	0.18	0.08	0.08	74.82	45.96	70.84	4.78	41.86
Ind D	17	64.52	33.56	12.56	50.36	25.12	10.70	23.60	7.92
Panel B: Developed Markets ex U.S.									
Unadj X	9	0.00	0.00	52.16	1.50	0.00	79.72	76.76	28.46
Ind and adj X	17	0.00	0.04	0.06	0.00	0.00	0.02	0.16	0.00
Ind X	9	0.12	39.22	52.28	23.68	99.12	59.60	14.94	36.72
Ind D	17	45.12	34.48	15.54	4.74	10.10	1.32	0.66	69.64
Cty and adj X	17	2.26	0.40	0.50	0.18	0.12	0.72	0.66	0.50
Cty X	9	32.46	21.32	43.78	82.64	38.86	56.94	89.32	78.78
Cty D	15	80.06	42.64	54.12	41.22	45.16	37.78	22.46	87.74
Cty, ind, and adj X	25	1.72	0.00	0.00	0.02	0.02	0.02	0.42	0.22
Cty and ind X	17	7.92	1.32	1.02	0.78	15.12	18.36	66.00	59.56
Cty and ind D	31	82.70	54.60	37.16	34.36	19.30	13.34	9.42	3.60
Panel C: Emerging Markets									
Unadj X	9	0.34	4.60	23.36	3.16	0.08	71.18	0.28	29.70
Ind and adj X	17	0.36	3.42	36.14	13.12	8.70	5.44	16.90	19.38
Ind X	9	6.90	23.22	0.42	95.56	89.66	65.28	68.40	36.64
Ind D	17	70.92	46.82	11.04	98.78	79.38	38.26	44.50	36.38
Cty and adj X	17	17.78	6.10	3.00	1.02	0.20	0.08	0.00	0.00
Cty X	9	74.36	75.78	71.60	63.12	53.90	57.20	98.24	27.22
Cty D	19	96.14	98.74	97.30	93.40	85.80	81.80	73.64	95.92
Cty, ind, and adj X	25	13.60	5.00	1.72	0.42	0.02	0.00	0.00	0.00
Cty and ind X	17	46.82	54.62	35.52	21.46	7.90	9.48	3.72	2.82
Cty and ind D	35	98.28	99.42	98.22	95.70	90.88	88.10	81.00	92.92
Panel D: World									
Unadj X	9	0.00	0.00	59.56	53.20	54.96	23.38	5.06	0.98
Ind and adj X	17	0.00	0.00	19.98	2.16	0.12	0.06	0.22	0.80
Ind X	9	0.48	0.98	7.52	0.54	35.74	5.62	4.20	10.88
Ind D	17	45.64	21.20	6.60	2.36	0.44	0.04	44.92	68.34
Cty and adj X	17	0.30	0.02	0.04	0.32	0.00	0.00	0.00	0.26
Cty X	9	30.84	12.52	64.86	11.60	2.26	19.58	43.96	88.96
Cty D	35	6.70	7.24	3.16	7.90	4.52	76.94	41.84	35.90
Cty, ind, and adj X	25	0.20	0.00	0.00	0.06	0.00	0.00	0.00	0.00
Cty and ind X	17	9.70	3.68	13.08	13.44	7.42	2.10	2.16	27.64
Cty and ind D	51	6.98	10.96	2.64	0.62	15.68	17.50	43.84	38.60

We report the p -value (in %) for the null hypothesis that the α are not driven by the instruments, $H_0 : A'A = 0$. Models are estimated on all stocks in the U.S. in Panel A, for which models with country effects are not estimated, in Developed Markets excluding the U.S. in Panel B, in Emerging Markets in Panel C, and in all countries in Panel D. Models differ by the instrument used in the specification of the α s (see Eq. (3)) and β s (see Eq. (4)). Model “Unadj X” uses unadjusted characteristics. Model “Ind and adj X” uses industry and industry-adjusted characteristic components, while Model “Ind X” uses only industry components. Model “Ind D” uses industry dummies. Model “Cty and adj X” uses country and country-adjusted characteristic components, while Model “Cty X” uses only country components. Model “Cty D” uses country dummies. Model “Cty, ind, and adj X” uses country, industry, and country- and industry-adjusted characteristic components, while Model “Cty and ind X” uses country and industry components. Finally Model “Cty and ind D” uses country and industry dummies.

5.4 Table 4: Model performance – Asset pricing tests

Read: International models with adjusted components often reject $H_0 : A'A = 0$, even with many factors. U.S. unadjusted-characteristic alphas become insignificant for larger K .

Economic message. The familiar U.S. result that characteristic alphas disappear once betas are modeled does not fully hold internationally.

5.5 Table 5 and Table 6: Comparison to other factor models; Significance tests for parameter groups

Read: Table 5 compares the component model with HR/GK and BHZ benchmarks. Table 6 shows that adjusted components are significant for alpha parameters and country components are significant for beta parameters.

Economic message. The mechanism is clear: AdjX explains alpha, CtyX explains risk exposure.

Table 5
Comparison to other factor models.

Model	DM ex U.S.					EM					World				
	K	Total R^2	$\Delta\%$	Pred. R^2	$\Delta\%$	K	Total R^2	$\Delta\%$	Pred. R^2	$\Delta\%$	K	Total R^2	$\Delta\%$	Pred. R^2	$\Delta\%$
Cty, ind, and adj X	7	16.76		0.42		7	23.27		0.78		7	15.74		0.60	
Panel A: Heston and Rouwenhorst (1994) and Griffin and Karolyi (1998)															
Equal-weighted	33	17.31	3	0.19	-55	37	27.07	16	0.52	-33	53	19.03	21	0.41	-32
Panel B: Bekaert et al. (2009)															
WLFF - CI	12	10.66	-36	-		12	17.30	-26	-		21	9.05	-43	-	
WLAPT - CI	12	13.44	-20	-		12	21.65	-7	-		21	15.49	-2	-	
WLFF - CS	12	14.34	-14	-		12	19.82	-15	-		21	13.37	-15	-	
WLAPT - CS	12	15.73	-6	-		12	24.58	6	-		21	18.74	19	-	

We compare the number of factors, Total R^2 s (see Eq. (5)) and Predictive R^2 s (see Eq. (6)), in percent, between Model "Cty, ind, and adj X" and benchmark models. We report percentage differences between benchmark model R^2 s and those of "Cty, ind, and adj X". Panel A reports on country-industry dummy model in which stock returns have a beta of one on a common factor and their respective country and industry factors. Panel B reports on the World-Local Fama-French (WLFF) and World-Local APT (WLAPT) models of Bekaert et al. (2009). We assign the portfolio-level betas to individual stocks based on either the country-industry portfolio set (CI) or the country-style portfolio set (CS). We use a 17-industry classification. All data are monthly, in USD, and end in

(a) Table 5: Comparison to other factor models.

Table 6
Significance tests for parameter groups.

Region	Characteristic components		
	Country	Industry	Adjusted
Panel A: p -values for the joint significance of A parameters			
U.S.	-	0.18	0.00
DM ex U.S.	73.62	9.72	0.00
EM	22.68	3.38	0.00
World	4.72	2.14	0.00
Panel B: p -values for the joint significance of B parameters			
U.S.	-	61.38	0.00
DM ex U.S.	6.20	15.42	0.00
EM	0.00	83.38	0.30
World	0.00	30.88	0.00

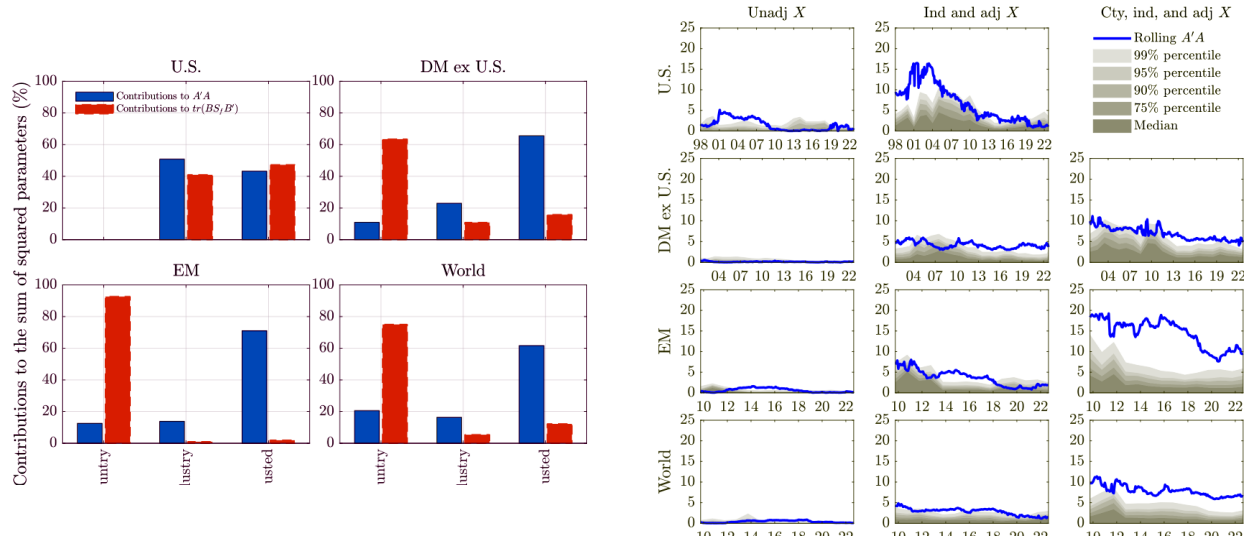
We report p -values (in %) for significance tests for parameter groups corresponding to characteristic components. In Panel A, we test the null hypothesis, $H_0: A'A_c = 0$, where A_c is the sub-vector of coefficients in A that correspond to country components. Similarly, we test for the significance of industry and adjusted component parameters using the test statistics $A'A_i$ and $A'A_a$, respectively. In Panel B, we test the significance of parameter groups in B across factors using test statistics $\sum_{k=1}^K B_{c,k}B_{c,k}'$, $\sum_{k=1}^K B_{i,k}B_{i,k}'$, and $\sum_{k=1}^K B_{a,k}B_{a,k}'$. In all cases, we use Model "Cty, ind, and adj X" with $K = 7$ factors. The p -values are obtained by bootstrapping under the null hypothesis. We use a 17-industry classification. All data are

(b) Table 6: Significance tests for parameter groups.

5.6 Figure 1: What is the relative importance of each characteristic component for alphas and betas?

Read: Adjusted components dominate alpha contributions in international markets, while country components dominate beta contributions.

Economic message. The same characteristic has different economic roles depending on its component: adjusted variation is alpha-relevant; country variation is risk-relevant.



5.7 Figure 2: Are alphas significant over time?

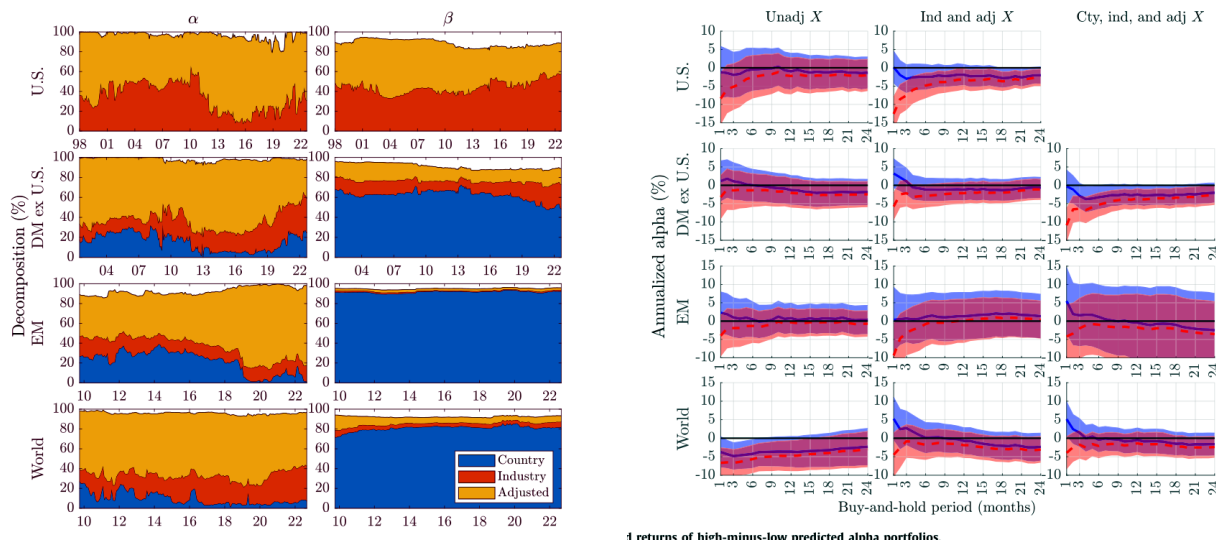
Read: Rolling alpha tests show that alpha significance declines over time but remains meaningful in some international component specifications.

Economic message. Alpha opportunities are time-varying and weakening, not stable arbitrage constants.

5.8 Figure 3: Component contributions to alphas and beta over time

Read: Adjusted components dominate alpha contributions in international markets; country components dominate beta contributions, especially in EM.

Economic message. The alpha-risk decomposition is persistent over time.



5.9 Figure 4: Risk-adjusted returns of high-minus-low predicted alpha portfolios

Read: Long-short predicted-alpha strategies do not reliably produce significant net returns after costs.

Economic message. Statistical alphas are not necessarily tradable alphas.

5.10 Figure 5: Risk-adjusted returns of high predicted alpha portfolios

Read: Long-only strategies avoid shorting costs but still generally fail to produce significant risk-adjusted performance.

Economic message. The long-only evidence reinforces the practical limits of alpha extraction.

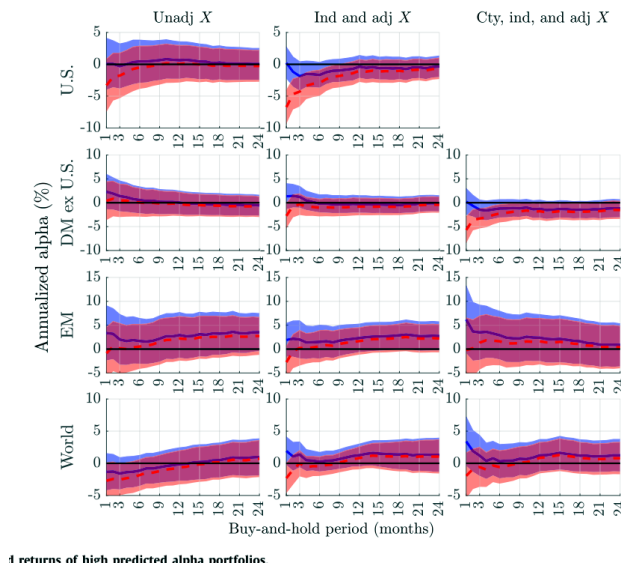


Figure 5: Risk-adjusted returns of high predicted alpha portfolios.

6 Short summary

- Characteristics contain country, industry, and adjusted components.
- AdjX is the main source of alphas in international markets.
- CtyX is the main source of comovement and beta exposure.
- Component-based IPCA models outperform unadjusted-characteristic and many dummy-based alternatives.
- Alpha significance declines over time and is difficult to monetize after transaction costs.

7 Conclusion

7.1 Core conclusion

Langlois shows that what matters in a characteristic depends on the asset-pricing object. Adjusted components matter most for alphas, while country components matter most for beta exposures and comovement.

7.2 Economic intuition

A raw characteristic mixes country-wide, industry-wide, and firm-specific information. Country-level information clusters stocks into common risk exposures. Within-country and within-industry residual variation is more closely tied to expected-return anomalies.

7.3 Implications for diversification

The results reinforce the importance of country structure for international diversification. Country characteristic components help explain why stocks comove within countries.

7.4 Implications for anomaly investing

Statistical alpha is not the same as tradable alpha. Estimation risk, parameter instability, turnover, and transaction costs weaken predicted-alpha strategies.

7.5 Limitations and future research

The decomposition is not causal and depends on the chosen characteristic set and country/industry taxonomy. Future work can connect adjusted-component alpha to limits to arbitrage, information frictions, and market segmentation.

7.6 Takeaway

The paper turns a broad characteristic signal into interpretable pieces: AdjX is mostly alpha-relevant, CtyX is mostly risk-relevant, and IndX is less dominant but still useful in some settings.